


It is interesting to note that the experiment of Hertz confirmed that light is an electromagnetic wave. But the same experiment also produced the first evidence for particle nature of light.

### Hallwachs' observation

In 1888, Wilhelm Hallwachs, a German physicist, confirmed that the strange behaviour of the spark is due to the action of ultraviolet light with his simple experiment.

A clean circular plate of zinc is mounted on an insulating stand and is attached to a gold leaf electroscope by a wire. When the uncharged zinc plate is irradiated by ultraviolet light from an arc lamp, it becomes positively charged and the leaves will open as shown in Figure 8.6(a).

Further, if the negatively charged zinc plate is exposed to ultraviolet light, the leaves will come closer as the charges leaked away quickly (Figure 8.6(b)). If the plate is positively charged, it becomes more positive upon UV rays irradiation and the leaves open further (Figure 8.6(c)). From these observations, it was concluded that negatively charged electrons were emitted from the zinc plate under the action of ultraviolet light.

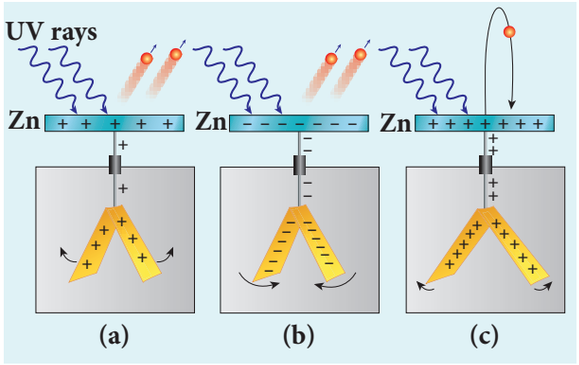



**Figure 8.6** Irradiation of ultraviolet light on (a) uncharged zinc plate (b) negatively charged plate (c) positively charged plate

### Lenard's observation

In 1902, Lenard studied this electron emission phenomenon in detail. His simple experimental setup is shown in Figure 8.7. The apparatus consists of two metallic plates *A* and *C* placed in an evacuated quartz bulb. The galvanometer *G* and battery *B* are connected in the circuit.

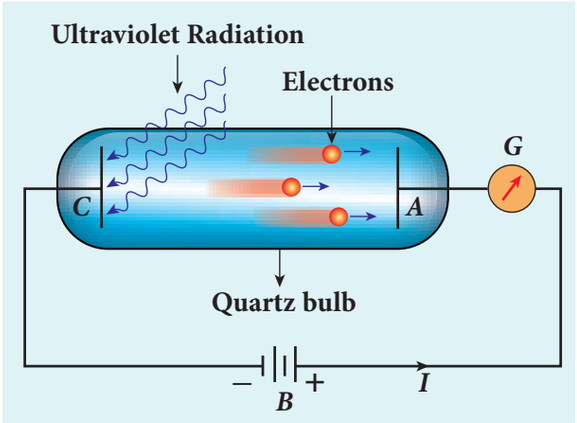



**Figure 8.7** Experimental setup of Lenard

When ultraviolet light is incident on the negative plate *C*, an electric current flows in the circuit that is indicated by the deflection in the galvanometer. On other hand, if the positive plate is irradiated by the ultraviolet light, no current is observed in the circuit.

From these observations, it is concluded that when ultraviolet light falls on the negative plate, electrons are ejected from it which are attracted by the positive plate *A*. On reaching the positive plate through the evacuated bulb, the circuit is completed and the current flows in it. Thus, the ultraviolet light falling on the negative plate causes the electron emission from the surface of the plate.

### Photoelectric effect

The ejection of electrons from a metal plate when illuminated by light or any other electromagnetic radiation



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